

Yazhou Zhu

Ph.D. in Computer Science and Technology | Medical Image Analysis and AI

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Professional Summary

Ph.D. in Computer Science and Technology specializing in medical image analysis under data scarcity and domain shift. My research trajectory spans unsupervised MR image denoising and 3D brain tumor segmentation, followed by few-shot prototype learning, cross-domain semantic matching, domain-robust optimization, uncertainty-aware prompting, and SAM adaptation. Google Scholar lists 16 publications, including 10 first-author papers; as of 7 September 2026, these works have received 333 citations, with an h-index of 9 and an i10-index of 8. My work covers segmentation, enhancement, and reconstruction in MR and CT, with future directions in medical foundation-model adaptation, multi-center generalization, and trustworthy medical imaging AI.

16	10	333	9 / 8
Scholar-listed works	First-author papers	Total citations	h-index / i10-index

Research Interests

Cross-domain Few-Shot Medical Image Segmentation: Prototype learning; label-efficient segmentation

Medical Foundation Model Adaptation: SAM and DINOv2 prompting; training-free adaptation; parameter-efficient adaptation

Medical Image Reconstruction and Enhancement: MR denoising; undersampled k-space reconstruction; self-supervised image priors

Trustworthy Multi-center Medical AI: Domain generalization; uncertainty estimation; robustness across modalities, sequences, devices, and institutions

Education

Ph.D. Computer Science and Technology Nanjing University of Science and Technology School of Computer Science and Engineering Advisor: Prof. Haofeng Zhang Cross-domain few-shot medical image segmentation; medical image reconstruction and enhancement	Sep 2022 - Jun 2026
M.Eng. Software Engineering Jiangnan University School of Artificial Intelligence and Computer Science Advisor: Xiang Pan Deep-learning-based MR image denoising and brain tumor segmentation	Sep 2018 - Jun 2021
B.Eng. Mechanical Design, Manufacturing and Automation Changzhou University GPA: 4.1 / 5.0 82 / 100	Sep 2014 - Jun 2018

Honors and Awards

2019 Second-Class Graduate Academic Scholarship, Jiangnan University	2017 Third-Class Scholarship, Changzhou University
2016 Third Prize, GstarCAD Cup East China CAD Design Competition	2016 Second-Class Scholarship, Changzhou University

Research Trajectory

Stage	Research Focus
Image Reconstruction and Enhancement	Began with unsupervised MR denoising and deep image priors, building experience in medical imaging inverse problems, structure preservation, and 3D data processing.
Few-Shot Segmentation	Developed prototype generation, debiasing, regional representation, and Transformer interaction methods for MR and CT segmentation with limited annotations.
Cross-Domain and Foundation Model Adaptation	Extended the research to cross-modality, cross-sequence, and cross-institution shifts, adapting SAM and DINOv2 to medical image segmentation through uncertainty-aware prompting.

Publications

Verified against the Google Scholar profile: 16 works in total. Citation counts in brackets are current as of 7 September 2026. Yazhou Zhu is the applicant.

First-Author Publications 10

1. **Yazhou Zhu**, Shidong Wang, Tong Xin, Haofeng Zhang. *Adversarial Prototypical Perturbation for Cross-domain Few-shot Medical Image Segmentation*. **ACM Transactions on Multimedia Computing, Communications, and Applications**, 22(8), Article 234, 1-27, 2026. [GS citations 0] [Link](#)
2. **Yazhou Zhu**, Shidong Wang, Tao Zhou, Zechao Li, Haofeng Zhang, Ling Shao. *Cross-Domain Few-Shot Medical Image Segmentation via Dynamic Semantic Matching*. **IEEE Transactions on Image Processing**, 34, 6669-6682, 2025. [GS citations 11] [Link](#)
3. **Yazhou Zhu**, Haofeng Zhang. *MAUP: Training-Free Multi-center Adaptive Uncertainty-Aware Prompting for Cross-Domain Few-Shot Medical Image Segmentation*. **MICCAI, LNCS 15966, 326-336, 2025**. [GS citations 12] [Link](#)
4. **Yazhou Zhu**, Minxian Li, Qiaolin Ye, Shidong Wang, Tong Xin, Haofeng Zhang. *RobustEMD: Domain Robust Matching for Cross-domain Few-shot Medical Image Segmentation*. **Artificial Intelligence in Medicine**, 167, 103197, 2025. [GS citations 8] [Link](#)
5. **Yazhou Zhu**, Shidong Wang, Tong Xin, Zheng Zhang, Haofeng Zhang. *Partition-A-Medical-Image: Extracting Multiple Representative Subregions for Few-Shot Medical Image Segmentation*. **IEEE Transactions on Instrumentation and Measurement**, 73, 1-12, 2024. [GS citations 51] [Link](#)
6. **Yazhou Zhu**, Ziming Cheng, Shidong Wang, Haofeng Zhang. *Learning De-biased Prototypes for Few-Shot Medical Image Segmentation*. **Pattern Recognition Letters**, 183, 71-77, 2024. [GS citations 21] [Link](#)
7. **Yazhou Zhu**, Shidong Wang, Tong Xin, Haofeng Zhang. *Few-Shot Medical Image Segmentation via a Region-Enhanced Prototypical Transformer*. **MICCAI, LNCS 14223, 271-280, 2023**. [GS citations 87] [Link](#)
8. **Yazhou Zhu**, Xiang Pan, Tianxu Lv, Yuan Liu, Lihua Li. *DESN: An Unsupervised MR Image Denoising Network with Deep Image Prior*. **Theoretical Computer Science**, 880, 97-110, 2021. [GS citations 27] [Link](#)
9. **Yazhou Zhu**, Xiang Pan, Jing Zhu, Lihua Li, Yuan Liu. *Denoising of Magnetic Resonance Images with Deep Neural Regularizer Driven by Image Prior*. **IEEE DSAA**, 255-263, 2020. [GS citations 5] [Link](#)
10. **Yazhou Zhu**, Xiang Pan, Jing Zhu, Lihua Li. *Multi-scale Strategy Based 3D Dual-Encoder Brain Tumor Segmentation Network with Attention Mechanism*. **IEEE BIBM**, 952-957, 2020. [GS citations 9] [Link](#)

Co-authored Publications 6

11. Bin Xu, **Yazhou Zhu**, Haofeng Zhang. *PSP: Harnessing Position and Shape Priors for Cross-Domain Few-Shot Medical Image Segmentation*. **MICCAI, 2026**. [GS citations 0] [Link](#)

Co-authored Publications continued

12. Yuntian Bo, **Yazhou Zhu**, Piotr Koniusz, Haofeng Zhang. *Focus on Background: Exploring SAM's Potential in Few-Shot Medical Image Segmentation with Background-centric Prompting*. **IEEE/CVF CVPR**, 30032-30041, 2026. [GS citations 2] [Link](#)

13. Bin Xu, **Yazhou Zhu**, Shidong Wang, Yang Long, Haofeng Zhang. *Few-Shot Medical Image Segmentation via Boundary-Extended Prototypes and Momentum Inference*. **Computer Vision and Image Understanding**, 263, 104571, 2026. Google Scholar lists the online publication year as 2025. [GS citations 3] [Link](#)

14. Ziming Cheng, **Yazhou Zhu**, Shidong Wang, Tong Xin, Haofeng Zhang. *Cascaded Alternating Refinement Transformer for Few-Shot Medical Image Segmentation*. **ACM Transactions on Intelligent Systems and Technology**, 2025. [GS citations 5] [Link](#)

15. Yuntian Bo, **Yazhou Zhu**, Lunbo Li, Haofeng Zhang. *FAMNet: Frequency-Aware Matching Network for Cross-Domain Few-Shot Medical Image Segmentation*. **AAAI**, 39(2), 1889-1897, 2025. [GS citations 73] [Link](#)

16. Tianxu Lv, Xiang Pan, **Yazhou Zhu**, Lihua Li. *Unsupervised Medical Images Denoising via Graph Attention Dual Adversarial Network*. **Applied Intelligence**, 51(6), 4094-4105, 2021. [GS citations 19] [Link](#)

Research Plan and Future Directions

- 1. **Low-label adaptation of medical foundation models:** Building on prior work with SAM and DINOv2 prompting, investigate training-free prompting, prompt tuning, and parameter-efficient fine-tuning so that general-purpose vision or vision-language foundation models can perform medical segmentation and recognition with limited pixel-level labels, image-level labels, or no target-domain training data.
- 2. **Multi-center generalization and trustworthy evaluation:** Study domain-robust representations, test-time adaptation, uncertainty calibration, and out-of-distribution detection under cross-modality, cross-sequence, cross-device, and cross-institution shifts, and establish interpretable multi-center evaluation protocols.
- 3. **Joint reconstruction, enhancement, and downstream analysis:** Jointly model MR reconstruction and denoising with downstream tasks such as segmentation and detection. Combine self-supervised learning, imaging physics priors, and structure-preserving constraints to improve diagnostic information fidelity under undersampling and low signal-to-noise conditions.
- 4. **Multimodal medical artificial intelligence:** Integrate MR and CT imaging with text reports and structured clinical data, focusing on cross-modal alignment, knowledge transfer, and reliable reasoning in data-scarce settings.

Technical and Research Skills

Category	Competencies
Research Methods	Few-shot and meta-learning; prototype learning; cross-domain generalization; adversarial learning; Transformers; uncertainty-aware prompting; foundation-model adaptation
Medical Imaging	MRI and CT; 2D and 3D segmentation; MR denoising and reconstruction; undersampled k-space; multi-center image analysis
Programming and Frameworks	Python; PyTorch; TensorFlow; C and C#; MATLAB
Data and Imaging Tools	SimpleITK; NiBabel; NumPy; Z-score normalization; cropping; augmentation; medical image preprocessing and analysis
Research Communication	Literature review in English and Chinese; academic writing in English; cross-institutional collaboration; independent management of algorithm development, experiments, and research outputs